

α -Fetoprotein (AFP) Tetra Profile

Test Details

METHODOLOGY

Chemiluminescent immunoassay

RESULT TURNAROUND TIME

2 - 5 days

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Test Includes

α -Fetoprotein, serum; dimeric inhibin A (DIA); human chorionic gonadotropin (hCG); unconjugated estriol (uE3)

Use

Screening test for open neural tube defects (detects 80% open spina bifida, 90% anencephaly), Down syndrome (detects 75% to 80%), and trisomy 18 (detects 73%)

Special Instructions

The following information must be provided: gestational age, date on which the patient was the stated gestational age, how gestational age was determined (LMP, EDD, US), patient's weight, patient's date of birth, patient's race (white, black, other), and insulin-dependent diabetic status. Also indicate relevant patient history (eg, prior neural tube defects, Down syndrome, ultrasound anomalies, or previous maternal serum screening specimen during this pregnancy). Complete information is necessary to interpret the test. Patient information may be provided to the laboratory using The Maternal Prenatal Screening requisition form 0900. Specimens must be collected before amniocentesis.

Down syndrome screening is offered for gestational ages 15.0 to 21.9 weeks. Open spina bifida screening is offered for gestational ages 15.0 to 23.9 weeks. The optimal gestational age for open spina bifida screening is 16.0 to 18.9 weeks.

Limitations

AFP tetra is a screening test. A positive AFP tetra result means that more diagnostic testing may be offered to the pregnant woman to determine if a neural tube defect or chromosome abnormality is present.

This test was developed and its performance characteristics determined by Labcorp. It has not been cleared or approved by the Food and Drug Administration.

Specimen Requirements

Specimen

Serum

Volume

5 mL

Minimum Volume

3 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Gel-barrier tube

Collection Instructions

Collect in serum separator tube with gel barrier. Allow blood to clot, avoiding hemolysis. Separate serum from cells by centrifugation.

Transport spun tube to testing laboratory.

Pour-off is not advised. Maternal serum specimens must be drawn prior to amniocentesis to avoid contamination with fetal blood.

Stability Requirements

Temperature	Period
Room temperature	7 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Storage Instructions

Room temperature

Causes for Rejection

Gross hemolysis; gross lipemia; quantity not sufficient for analysis;
improper specimen type

References

Haddow JE, Palomaki GE, Knight GJ, Foster DL, Neveux LM. Second trimester screening for Down's syndrome using maternal serum dimeric inhibin A. *J Med Screen*. 1998; 5(3):115-119. PubMed 9795869

(<http://www.ncbi.nlm.nih.gov/pubmed/9795869>)